

Revenue Choice on a Serial Network

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ABSTRACT

A model to examine the choice by jurisdiction whether to finance roads with taxes or tolls is developed. The idea of decentralized, local control and multiple jurisdictions distinguishes this analysis from one where a central authority maximizes global welfare. Key factors posited to explain the choice include the length of trips using the roads, the size of the governing jurisdiction, the elasticity of demand to revenue instruments, and the transaction costs of collection - which dictate the size and scope of the free rider problem associated with financing. Spatial complexity in this problem results from the fact that jurisdiction residents use both local and non-local networks, and each jurisdiction's network is used by both local and non-local residents. The central thesis argues that, since jurisdictions try to do well by their residents who are both voters and travelers, the effects of a revenue instrument on local residents is a key consideration in the choice of that revenue instrument. Decentralization of control and lower toll collection costs are identified as conditions under which tolls would more likely become the preferred revenue instrument for highways.

1. INTRODUCTION

Cordon tolls are becoming a popular method of restricting traffic in cities and financing new infrastructure. Singapore, Oslo, Trondheim, and Bergen are cities with such a mechanism in place. Imperfect cordons, such as a toll barrier on a major highway with toll-free entrance and exit ramps, have traditionally been used both in the early days of turnpikes and more recently on some limited access highway systems. In both cases, local trips (which remain within the cordon) do not pay tolls, while through trips (which cross the cordon) do. On the other hand, tax financing has been common both historically (initially the tax was in terms of labor), and more recently for local roads, which are funded through general revenues. Taxes also rely on a cordon, the boundary of the relevant jurisdiction, within which they are assessed.

In this paper a model of network financing is developed which incorporates the basic features of the economic structure of transportation networks. It includes the demand and supply interaction, the choices available to actors (consumers and producers), and the linkage between the two when the network is locally owned by the residents of its jurisdiction. The idea of decentralized, local control and multiple jurisdictions distinguishes this analysis from one where a central authority maximizes global welfare. The model's theoretical results should be consistent with what is empirically known about network financing. It should thus describe what network financing choices are made under various circumstances. Policies can then be drafted which alter circumstances to affect the desired choice of financing mechanism.

The main complication is the joint production and consumption of the key good (network services) by the jurisdiction and its residents. The network operator (jurisdiction) makes the network available while residents consume the network for traveling. Spatial complexity in this problem results from the fact that jurisdiction residents use both local and non-local networks, and each jurisdiction's network is used by both local and non-local residents. The network is not perfectly competitive and thus retains some monopoly power. The degree of locality in the use of the network directly shapes the local welfare resulting from a particular revenue mechanism, and itself is a function of jurisdiction size. The choice of financing instrument must trade off the number of spatial free riders, system users who don't pay their cost because of the location, and the costs of collection. The price charged for a given instrument is limited by the elasticity of demand on those who are charged.

The central thesis argues that, since jurisdictions try to do well by their residents who are both voters and travelers, the effects of a revenue instrument on local residents is a key consideration in the choice of that revenue instrument. It is assumed that jurisdictions

responsible for network financing behave as if they have the objective of local welfare maximization. Local welfare here reflects the consumers' surplus of residents of the jurisdiction and the profits accruing to the locally controlled network authority that the jurisdiction owns and manages. Levinson (1997) considers the thesis from a historical perspective. This paper, drawing from Levinson (1998), approaches the questions analytically to develop a positive theory of the choice of revenue mechanism over space as a function of key factors.

It is posited that the choice of tax or toll (or the optimal combination of both), while being historically contingent, is a function of several ahistorical or economic factors. These include the length of trips, the size of the jurisdiction, the transactions costs of collecting revenue, demand, and the cost of providing infrastructure. These properties indicate the nature of the free rider problem under the two different revenue instruments (tax, toll). In the case of tax-only financing, travelers from outside the taxing jurisdiction do not pay taxes to support the construction, operation, and maintenance of the road. In the case of toll-only financing, travelers entering and exiting the network within the toll cordon pay nothing. A perfectly excludable toll cordon, which tracks everyone entering and exiting each link, is not costless, and cannot necessarily be implemented everywhere. The free rider problem will occur whenever there is an incomplete or uneven revenue instrument, that is when financing does not capture every user in proportion to use. In realistic situations of highway transportation, there are transaction costs involved in collecting revenue, which dictate that a complete and perfect revenue instrument is not available on every link. It is an assumption of this work that jurisdictions would rather place the burden of financing on non-local travelers in order to maximize local welfare. Therefore, in the case of local control there is a preference to tax through trips and exempt trips which both originate in and are destined for the jurisdiction (locals), thereby allowing locals to be the free rider. At the extreme case, that of a single short road segment (a near perfect cordon), this approaches the efficient revenue instrument of tolling nearly all travelers in proportion to use, as there are very few, if any, "locals". However if those who make through trips control the political process, or through trips are really "local" because the road authority has a broad geographical scope, then taxes may be preferred because of their lower transactions costs.

This hypothesis does not claim to be a total explanation under all circumstances, as a socio-political system like infrastructure financing has many influences. For instance, perfect excludability on roads coupled with the presence of "free" alternatives, reductions in toll collection costs, and private ownership, may all increase the willingness of a jurisdiction to tolerate tolls even on local residents. The socio-economic level of the

jurisdiction may influence its willingness to adopt congestion pricing. Further, the influence of key players in business and politics with specific preferences are unpredictable and may greatly shape the decisions made over time. Another factor is the prevailing ideology of government. In the 18th and 19th centuries, a philosophy of limited, decentralized government, or *laissez faire*, was conducive to private enterprise at all levels, including roads. However, this philosophy declined in America in the twentieth century, at least through the early 1970's. Finally, regional rivalry and the idea of progress certainly have their place, promoting one-upmanship and construction for the sake of construction.

The model presented in this paper provides a strategic framework for assessing the outcome of alternative revenue instruments. This paper examines the free rider problem in relation to transaction (toll collection) costs. This model explicitly compares the welfare associated with taxes and those associated with tolls on a simplified network with various jurisdiction sizes. The actions available to network operators and the posited objectives that the two main sets of actors, the network operator (owned by residents) and travelers, are presented. Then, the network geometry of this model is illustrated. The model of flow as a function of trip length and tolls is explained, as is the resultant consumers' surplus. Profit calculations, and their component cost and revenue equations are presented. The models are then evaluated using assumed parameters to better understand its implication. Finally, key points, some policy conclusions, and directions for future research are summarized.

2. ACTIONS AND OBJECTIVES

A jurisdiction is defined as the owner of the road authority responsible for maintaining the road. These jurisdictions, or network operators, have several classes of actions, this research focuses on selecting a revenue instrument (taxes or tolls) and price or rate (collectively called revenue mechanism). Broadly, the two available revenue instruments are taxes and tolls. As used here, a *toll* (denoted by the Tau symbol (t)) is a fixed sum of money charged for a specific service or privilege (for instance, the right to travel on a link or subnetwork). Our toll is a fee levied by the road authority as travelers cross a cordon. In contrast, the term *tax* (denoted by the Chi symbol (c)) is defined as a fixed sum of money charged for a general service or privilege, such as the support of government or roads in general, independent of use. Our tax is a periodic (e.g. annual) fee levied by a jurisdiction on its residents in the form of a poll tax. The distinction between a tax and a toll is in how specific or general the service is which is being provided and how closely it aligns with the revenue mechanism. Thus, a gas tax is more like a toll than a property, poll, or income tax is. Gas taxes are not analyzed in this model, as they require

the additional complication of determining where, in relation to the home, a driver purchases gasoline.

In general, it is assumed that jurisdictions have the objective of local welfare maximization ($\text{Max } W_L$), where welfare is defined narrowly as the sum of profit (loss) from administering the road and consumers' surplus for its residents excluding external costs.

$$\text{Max}_{P_i} W_L = P_i + U_i \quad (1)$$

where:

Variable	Definition
P_i	Producer's Surplus (Profit) on network owned by jurisdiction i
U_i	Consumers' Surplus (transportation and non-transportation) of residents of jurisdiction i
P_i	Vector describing price of infrastructure, a function of location of trip origin, destination, location of toll-booths, revenue mechanism (incl. rate of odometer tax, cordon toll, or perfect toll and the basis of that toll), detailed later in text

While the most realistic assumption if the network and the user are independent is that the network operator will maximize profit while users will maximize their own utility, in our model the residents own the network through the jurisdiction. So the jurisdiction which owns the network is comprised of residents who (collectively through their government) can set the policies for the network (revenue mechanism) to achieve a maximization of their own welfare. Because profits are (or can be) redistributed to local residents, revenue in excess of costs from local residents is returned to them. Therefore, treating a jurisdictions and its residents as a single block is not unreasonable. The objective of a jurisdiction managing a roadway to maximize welfare for its residents, in addition to its obligation to cover costs, follows a long tradition of research (Downs 1957).

3. NETWORK GEOMETRY

The network geometry analyzed here consists of an infinitely long road, as illustrated in Figure 1. There are two types of cordons along the road: *jurisdiction boundaries* and *toll-booths*. A *jurisdiction* covers an area that contains, owns, and operates a portion of the road and is located between jurisdiction boundaries. In our model, all jurisdictions are identical in their fundamental features (including size), so we select one for analysis, which is called the *jurisdiction of interest* (J_0). The jurisdiction of interest covers the portion of the road between boundary points a and b , so its size is

represented by the distance $|b-a|$. All other jurisdictions (to the east and west of J_0) are collectively called the *environment* (E). Jurisdictions to the west of J_0 are denoted by J_{-n} , where n denotes the number of jurisdictions away J_{-n} is from J_0 . Similarly jurisdictions to the east of J_0 are denoted by J_{+n} . In our analysis of cordon tolls, we will assume that all toll-booths are located on jurisdiction boundaries, but not that all jurisdiction boundaries necessarily have toll-booths. Each jurisdiction which has cordon toll-booths has them both at the entrance and at the exit of the jurisdiction.

This geometry can be directly interpreted in an intercity context. The geometry can represent one jurisdiction (among many) which has authority over its portion of the long road, such as along an interstate highway under the authority of multiple states. In a more general problem, different densities of trip origins and destinations along the road could represent the center and periphery of an urban region.

In order to make the analysis more convenient, *sections* (S) will be defined as aggregations of jurisdictions. For simplicity in our analysis, three sections are defined: the area comprised of all jurisdictions west of the jurisdiction of interest J_0 $\{J_{-1}, J_{-2} \dots \in S_{-}\}$, the jurisdiction of interest $\{J_0\}$, and the area comprised of all jurisdictions east of J_0 $\{J_{+1}, J_{+2} \dots \in S_{+}\}$.

User *classes* (G) are defined as section-to-section (rather than jurisdiction-to-jurisdiction) interactions. In principle the number of section-to-section interactions (classes) is S^2 , where S is the number of sections. This chapter analyzes a one-way road, but assumes underlying symmetric demand functions, and round trips. On a one-way road, the number of relevant section-to-section interactions is reduced from the number S^2 , as trips can't exit upstream of where they enter. Furthermore, trips which do not travel through J_0 can be eliminated from consideration. Therefore, on a one-way road the three sections define only four section-to-section interactions, our *classes of interest*. The user classes are shown in Table 1.

4. CONSUMPTION

4.1 Demand

Flow ($f(z)$) across any point (z) on a road can be described by the function below, following and extending Newell (1980):

$$f(z) = \sum_{x < z} \sum_{y > z} (P_T(x, y; P_I(x, y))) dx dy \quad (2)$$

where:

Variable	Definition
$f(z)$	flow past point z
$r(P_T(x,y;P_I(x,y)))dxdy$	demand function representing the number of trips that enter facility between x and $x + dx$ and leave between y and $y + dy$
$P_T(x,y; P_I(x,y))$	generalized cost of travel to users (defined below)
x, y	where trip enters, exits road
z	point on road
$P_I(x,y)$	Price of infrastructure as function of location

A key assumption is that markets are non-substitutable. This means that there is no cross-elasticity of demand. For instance, trips remaining entirely within J_0 (class G_{00}) are unaffected by price changes by J_{+1} . There remain supply side effects, so that a change in price by J_{+1} which affects the demand for trips using roads in both J_0 and J_{+1} (such as G_{0+} trips) may affect prices faced by travelers in J_0 (say G_{-0} trips), even if they do not travel in J_{+1} . This price-demand interdependence can be explained by the following chain of logic: First the optimal tolls in any jurisdiction, including J_0 , depend on the demand function for the link; second, the demand on the link in J_0 depends on the demand of all trip classes using that link; and third those trip classes that use links in more than one jurisdiction - depend on prices on the links in each jurisdiction.

The argument of the demand function is assumed to be a weighted sum of the time and money costs of travel. A negative exponential form is used, therefore we may rewrite the density function as dependent on the total price users pay (P_T), a decay coefficient (a), as well as a multiplier (d) representing the number of trips generated per unit length.

$$r(P_T(x,y; P_I(x,y))) = de^{aP_T(x,y; P_I(x,y))} \quad (3)$$

where:

Variable	Definition
a	decay coefficients: $a < 0$
d	multiplier, number of trips per unit length $d > 0$
r()	demand function
P_T	generalized cost of travel
$P_I(x,y)$	price of infrastructure as function of location
x, y	where trip enters, exits road x<y (to account for the fact that it is a one way road)

The total price users pay for travel (P_T), is the sum of several components. Direct infrastructure charges (P_I) transferred to the network operators depend on the revenue policy selected by each jurisdiction (cordon tolls or general taxes), and will be discussed in more depth in a subsequent section.

Private vehicle costs (P_V), depend on the distance traveled ($|y-x|$), the cost per unit distance (n) as well as a number of fixed components (z) depending on the age and type of vehicle used.

Freeflow travel time costs (P_F) depend on trip length ($|y-x|$) and freeflow speed (S_F), as well as the value of time (V_T). Congested travel time is not dealt with here. Implicit in this model is that jurisdictions have the obligation of maintaining a level of service with a resulting freeflow speed which is consistent with congested speeds. Thus “congestion effects” are ascribed to infrastructure costs which are proportional to traffic flow (described in the cost section).

To simplify the analysis we assume no (dis)economies of scale and we assume smoothly and continuously increasing infrastructure costs. External costs are also excluded, since by definition they do not figure into the calculations of the network operator.

$$P_T = P_I + P_V + P_F = P_I(x, y) + (z + u|y - x|) + V_T \frac{|y - x|}{S_F} \quad (4)$$

where:

Variable	Definition
n	private variable cost (per unit length)
$P_I(x, y)$	price of infrastructure as function of location
P_T	generalized cost of travel
P_V	private vehicle costs
S_F	Freeflow speed
V_T	value of time
x, y	where trip enters, exits road
z	private fixed vehicle cost

4.2 Consumers' surplus

The sum of transportation consumers' surplus for all trips originating in jurisdiction J_0 (U_0) for the two relevant user classes ($U_{00} + U_{0+}$) is given by the following equation:

$$U_0 = U_{00} + U_{0+} = \int_{a-x}^b \int_{P_I(x, y)}^{\infty} r(P_T(x, y; p)) dp dy dx + \sum_{n=1}^b \int_a^{(n+1)(b-a)} \int_{P_I(x, y)}^{\infty} r(P_T(x, y; p)) dp dy dx \quad (5)$$

where:

Variable	Definition
U_0	consumer's surplus of trips originating in J_0
U_{00}, U_{0+}	consumer's surplus for trips of G_{00} and G_{0+}
$r(P_T(x, y; p)) dx dy$	demand function
$P_T(x, y; p)$	generalized cost of travel to users at price p
$P_I(x, y)$	price of infrastructure as function of location
n	index for number of toll-booths on road
a, b	jurisdiction J_0 cordon locations
p	user monetary cost (integrated from P_I to infinity)
d	spacing between toll-booths (e.g. $d=b-a$)

The rate of cordon toll that is assessed by jurisdictions on trips of the appropriate group multiplied by the basis over which the rate is assessed to compute the price paid (P_i). Local trips (G_{00}) don't pay cordon tolls (because they don't cross the toll cordon). Cordon tolls on locally originating, non-locally destined trips (G_{0+}) are assessed both by the home jurisdiction (J_0 assesses a toll rate r_l) and by other jurisdictions (which may assess a toll rate r_l). Because jurisdictions are identical, the effective price under identical policies (and toll-setting behavior) is identical across jurisdictions.

Technically speaking, we define each x-y pair as a distinct market (rather than defining each section-to-section pair as a distinct market). When we do so, then the x-y pair has its consumers' surplus measured before it is aggregated with other x-y pairs. This requires integrating over the range of tolls for each flow, and then integrating all the resulting consumers' surpluses over the relevant spaces. Fortunately, by Fubini's Theorem, so long as a function is continuous over real space, the order of integration of a double (or triple, etc.) integral does not matter. So the results would be the same, independent of how the markets are defined.

5. PRODUCTION

5.1 Profit

Profit or producers surplus (P_i) is defined below as total revenue from transportation (R_T) minus total (fixed and variable) costs (C_T):

$$P_i = R_T - C_T \quad (6)$$

where:

Variable	Definition
P_i	profit to jurisdiction i
R_T	total revenue from tolls
C_T	total cost

In our model, any loss is by definition made up from general taxes, and any profit is used to reduce them. Therefore the impact of general taxes to support roads on welfare can be measured in terms of the profit or loss of the network operator. Because general taxes and transportation demand are assumed to be independent, taxes don't need to be measured explicitly.

5.2 Cost

We take as a network cost model a function where total costs to the network operator (C_T) are an increasing function of jurisdiction size (C_S), traffic flow (C_r), and variable and fixed toll collections (C_{CV}, C_{CF}). We assume a linear model, that is there are no (dis-) economies of scale or scope. The first cost category, cost as a function of jurisdiction size, can be considered analogous to a fixed cost. It depends only on the size of the jurisdiction $|b-a|$ and the cost per unit distance (g) of constructing and operating infrastructure. The second cost category, cost as a function of traffic flow, depends on the vehicle distance traveled in the jurisdiction, which is multiplied by a cost per vehicle distance traveled (f). Because we are assuming that the network is sized to ensure a given (uncongested) level of service, the cost as a function of traffic flow is a composite of infrastructure capital and operating and maintenance costs. While the first cost category is determined by the length of the road, the second is determined in part by the width of the road necessary to ensure a particular level of service. The third and fourth cost categories are the variable and fixed costs of collecting tolls. The variable cost depends on the number of toll-booths (K_i) maintained by jurisdiction i and flow at each toll booth k , $f(k)$, as well as the cost per collection transaction (q). The fixed portion simply multiplies a fixed cost per toll booth (k) by the number of toll booths in the jurisdiction (which is 0 in the case of general taxes and 2 in the case of cordon tolls). The model can be expressed as below:

$$C_T = C_S + C_r + C_{CV} + C_{CF} = g|b - a| + f \int_a^b (z) dz + q \sum_{k=1}^{K_i} f(k) + kK_i \quad (7)$$

where:

Variable	Definition
C_T	total cost
C_S	fixed cost of jurisdiction size
C_r	variable cost of traffic flow
C_{CV}	variable cost of toll collection
C_{CF}	index for number of toll-booths on road
a, b	cordon boundary locations
g, f, q, k	cost coefficients
z	point on road

$f(z)$	traffic flow across point z
k	index of tollbooths in jurisdiction i
$f(k)$	traffic flow across tollbooth k
K_i	total number of tollbooths in jurisdiction i

5.3 Revenue

Revenue depends on the specific instrument chosen, the rate, and the quantity. The following table gives the equations for revenue. For instance the revenue from tolls (R_T) collected at toll-booths in jurisdiction i (K_i) ($k=1, \dots, K_i$) on the road is given below. In the case of jurisdiction based cordon tolls, $K_i=2$ (a and b).

$$R_T = r_t \sum_{k=1}^{K_i} f(k) \quad (8)$$

Variable	Definition
R_T	Total revenue (\$)
r_t	rate of toll (\$/crossing)
k	index of tollbooths in jurisdiction i
$f(k)$	traffic flow across tollbooth k
K_i	total number of tollbooths in jurisdiction i

6. PRICE OF INFRASTRUCTURE

6.1 General Structure

The price of infrastructure that a traveler faces depends on the policies of all jurisdictions. For clarity, we will consider two policies which jurisdiction J_0 imposes (tax or cordon toll), recall that a toll policy if it results in a loss may require taxes to cover costs, while if it generates profits will lower taxes (or result in negative taxes). We also need to determine the policy imposed by other jurisdiction. We assume two types of environments: all other jurisdictions (aside from J_0) impose taxes or all other jurisdictions impose cordon tolls.

In general, the price of infrastructure is a function of the rate of cordon toll (r_t) and the quantity over which each unit rate is applied (number of tolls crossed). The price of infrastructure can be decomposed into the price inside J_0 and the price outside J_0 , which are summed to attain the price paid by users. This decomposition enables a clear analysis of situations where a jurisdiction employs one policy while the environment imposes another.

6.2 Incidence of Tolls

Under a general tax policy, user payment for infrastructure is independent of the amount or location of travel. The rate of toll paid to a jurisdiction which imposes taxes is 0 for all user groups.

The price of infrastructure under cordon tolls deserves some discussion. A toll policy is more complicated than a general tax policy, as tolls affect demand while taxes don't. In the case of cordon tolls, direct infrastructure charges transferred to the network operators depend on the location of the origin and destination, that is an incidence matrix ($I(x,y,n)$) which represents whether toll-booth n is crossed for trips between x and y . Infrastructure charges also depend on the toll per crossing (r_{tn}), which is a policy variable available to the various jurisdictions. This can be represented as:

$$P_I(x,y) = \sum_{n=1}^{\hat{A}} r_{tn} [I(x,y,n)] \quad (9)$$

where:

Variable	Definition
$P_I(x,y)$	Price of infrastructure born by users
n	index of toll-booths along road
r_{tn}	rate of toll at toll crossing n
$I(x,y,n)$	toll incidence matrix: $I(x,y,n) = 1$ if $x < L(n) < y$; $I(x,y,n) = 0$ otherwise
$L(n)$	location of toll-booth n

When a cordon toll policy in J_0 is imposed, all boundary crossing trips entering and exiting J_0 (G_{+0} , G_{0-}) pay a toll of r_t at the cordon. Local trips (G_{00}) do not cross cordons,

and thus have $I(x,y,n) = 0$ for all n . Through trips (class G_{-+}), pay the toll of r_t at both the entry and the exit cordon, and so have a total payment of $2r_t$ in J_0 .

Similarly, the cordon toll environment is more complicated than the general tax environment and also warrants some discussion. When all other jurisdictions impose taxes on their own residents to finance their roads, we set external tolls to zero ($r_t=0$). However, when those other jurisdictions impose a cordon toll policy, the analysis includes tolls for all other jurisdictions (r_t). It is assumed, to simplify the calculations, that all jurisdictions are identically sized.

When either J_0 or other jurisdictions or both use tolls, then we must recognize that a certain fraction of boundary crossing and through trips pays tolls at more than one toll-booth. For instance, with a toll policy in J_0 and an all-toll environment, all G_{-+} trips must pay tolls at the exit of J_{-1} , the entrance of J_0 , the exit of J_0 , and the entrance of J_{+1} . Some of the longer trips must also pay at the exit of J_{+1} and entrance of J_{+2} . An even smaller fraction must also pay at the exit of J_{+2} and entrance of J_{+3} , and so on. This is illustrated in Figure 2. Similarly, though not shown, some trips enter the network before J_{-1} , fewer before J_{-2} , and so on.

At any cordon, a trip must pay the exit toll for the jurisdiction it is leaving and the entrance toll for the jurisdiction it is entering. So through trips (class G_{-+}) don't all pay the same toll. If tolls are spaced regularly, then under a policy of discrete cordon tolls the flow that through trips face can be represented by an infinite series of integrals. Similar reasoning can be applied to both classes of boundary crossing (G_{0+} and G_{0-}) trips.

6.3 Toll Setting

The discussion to date still leaves some latitude for solving the tactical problem of determining the rate of toll. The issue, in solving for the J_0 's toll (r_t), is what does J_0 assume the other jurisdiction use as a toll (r_t) when it is known what policy they choose. We employ a non-cooperative equilibrium, which assumes no collusion (implicit or otherwise) between jurisdictions. Thus we are seeking a non-cooperative Nash equilibrium for toll-setting. This means that J_0 can do no better by changing its toll given what all other jurisdictions do, while each other jurisdiction can also do no better. This does not necessarily result in the best satisfaction of the objective function, but is sustainable.

We use an iterative approach to solve for this equilibrium. For each iteration, we assume J_0 sets its payoff maximizing toll as if the tolls in other jurisdictions are fixed.

Under that constraint, the best assumption J_0 can make is that the other jurisdictions are using their last posted toll. Their last posted toll happens to be the toll previously solved for by J_0 , since all jurisdictions are identical, and simultaneously performing these calculations.

To translate this into an algorithm: during solution round (i), J_0 assumes that r_t is equal to J_0 's toll in solution round i-1: $r_t^i = r_t^{i-1}$. The algorithm says: given r_t and all the other pertinent variables, J_0 finds the welfare maximizing r_t , updates r_t , and solves until equilibrium ($r_t^* = r_t^*$).

7. MATHEMATICAL SOLUTION

These reduced forms are given in Tables 2-4, which are discussed below. With the coefficients and jurisdiction size fixed (while tolls remain variable), many of the terms become constants for purposes of analysis.

The general expression (for utility, cost, revenue) (Y) can be written in the form of:

$$Y^{PP} = g_G(r_t^{qq}, r_t^{qq}) b_{YG} \quad (10)$$

where: $g_G(r_t^{qq}, r_t^{qq})$ is a function of tolls and crossings associated with user groups (G).

$g_G = 1$ when Y represents a fixed cost (C_S, C_{CF}).
 $=$ equation given in Table 3 otherwise.

q, q denotes policy (tax or toll) of J_0 , other jurisdictions

r_t, r_t denotes rate of toll of J_0 , other jurisdictions

b_{YG} is the portion of the expression specific to the cost/revenue/consumers' surplus category, given in Table 2.

Table 2 displays the constant b_{YG} component, which is common to all four combinations of J_0 /environment policies, but differs for each cost/revenue/consumers' surplus category (Y) and user group (G). An attempt was made to further reduce this (that is to attain a simpler b_Y component, applicable across all user groups for a given component), but the simplification could only be achieved at the expense of constancy in the g_G component across user groups.

Table 3 develops the component associated with user groups (g_G). This component is complex because it captures the interface between the continuous demand functions and the discrete location of toll-booths. This component is formulated to be general as to

handle both tax and toll policies in both all-tax and all-cordon toll environments. The form of the g_G equation can be simplified in the absence of tolls in either J_0 or other jurisdictions or both, as then the respective tolls (r_t) or (r_l) can be set to 0.

Table 4 shows the underlying assumptions about the rate of toll in J_0 (r_t) and in the other jurisdictions (the environment) (r_l) under the various J_0 policy and environment conditions. By substituting the appropriate toll value into the g_G and b_{YG} equations, those equations can be solved for different policies in J_0 , (q) and in the environment E , (q).

Assuming the demand functions described, and keeping toll rates fixed, we may draw some basic conclusions in the case of cordon tolls. As the size of jurisdiction J_0 increases, that is as $|b-a|$ gets large:

1. While the total number of trips crossing cordon a remains the same or increases, the ratio of non-through (F_{-0}) to through (F_{+}) trips increases. As b gets farther from a , the number of trips crossing both a and b decreases, and the negative effect (finance externality) of a toll at b on traffic crossing a declines. Any trip going the distance $|y-x|$ is more likely to take place if it crosses one toll-booth rather than two.
2. Similarly, the total number of trips crossing cordon b remains the same or increases as the ratio of non-through (F_{0+}) to through (F_{+}) increases.
3. The total number of trips originating in or destined for jurisdiction J_0 (F_{00} , F_{0+} , and F_{-0}) increase. However, the number of non-local (F_{0+} and F_{-0}) trips approach a limit (the maximum flow past the cordon) while local (F_{00}) trips increase with jurisdiction size.

8. EVALUATION

The toll which maximizes welfare can be found by solving the welfare maximization problem, that is by setting the first derivative of the welfare expression to zero and solving for the toll. As can be seen from the slew of equations described above, this is a rather long expression in its full (expanded) form.

Perhaps a more comprehensible way to understand what is going on is by examining some illustrations of how welfare changes as certain factors are altered. The underlying assumptions are given in Table 5, developed from empirical studies of travel cost and demand (Levinson 1998), any changes to those assumptions are discussed in the text. It is assumed that all other jurisdictions are using tax financing, so that the financing of those jurisdictions does not effect demand.

Figure 3 shows how welfare changes in J_0 as it varies its tolls assuming all other jurisdictions employ a tax, keeping fixed jurisdiction size (at 10 km) and all other factors.

Tolls of zero are computed both with and without collection costs for comparison. Higher welfare is found without collection costs, a toll of value zero is equivalent to a tax policy in the absence of collection costs. Here, (with \$0.05 increments on the graph) the welfare maximizing toll can be read as \$0.70, lower than the profit maximizing toll (\$1.10). This reflects that consumers' surplus as well as profit are included in the welfare objective function. The serial (complementary) nature of the network creates interactions which don't necessarily exist in a strictly competitive market analysis. Welfare can remain positive despite the lack of cost-recovery, suggesting that under certain circumstances welfare maximization may result from a combination of cordon tolls and tax financing used to subsidize roads. At low tolls, costs exceed revenues (at very high tolls, costs exceed revenues as well). Here, however, the optimum toll rate generates a positive profit, which, as explained above, is assumed to be returned to the jurisdiction's residents through reduced taxes or a direct payout.

In the all-toll environment, the welfare and welfare maximizing toll of a given jurisdiction depend upon the tolls of other jurisdictions. We solve the tactical, toll-setting problem by assuming jurisdictions don't cooperate. This gives the Nash equilibrium toll, which is necessary to find the Nash equilibrium policy. At the Nash equilibrium toll, the tolls set by all jurisdictions will be the same. To find the Nash equilibrium value, we assume that all jurisdictions other than J_0 charge the same toll, and find the value for that toll such that J_0 's welfare maximizing toll takes the same value. Figure 4 plots consumers' surplus, profit, and total welfare of a jurisdiction against the toll it sets, assuming other jurisdictions are assessing the Nash equilibrium toll. Observe (by comparing Figures 3 and 4) that for any toll level, the welfare attained by J_0 is less when other jurisdictions are charging tolls. At low toll values, the welfare difference is dominated by consumers' surplus, reflecting the payments that J_0 's residents are making to other jurisdictions. At high toll levels, profit disparities predominate, since the tolls in other jurisdictions are suppressing lucrative toll crossings at J_0 . However the welfare maximizing toll in the all tax-environment is approximately the same (\$0.70) as that in the all-toll environment. This implies that the system is fairly stable, tolls in one jurisdiction will not fluctuate significantly as a result of road finance mechanism changes in other jurisdictions.

When the revenue instrument is taxes, it is fairly clear that to achieve a maximum of local (transportation and non-transportation) welfare while still recovering costs, the rate of taxes must be set such that total revenue equals total costs. The total welfare associated with a tax policy in an all-tax environment thus rises linearly with jurisdiction size. Because the environment and J_0 have identical policies, neither of which affect travel demand, there is no per capita variation in welfare, cost, or consumers' surplus. This is illustrated by the

horizontal line (denoted by “ $W(\text{Tax})/\text{Size}$ ”) in Figure 5, which displays the welfare resulting from both tax and toll policies in an all-tax environment.

Figure 5 also shows how welfare (W), profit (π), and consumers’ surplus (U) vary with jurisdiction size under a cordon toll policy in J_0 and an all-tax environment. Total welfare increases simply because the number of trips (and thus consumers’ surplus) increases. Per capita transportation consumers’ surplus increases with size as the effective price falls (more and more trips are local and thus pay no toll). This happens because the proportion of non-toll paying local trips increases relative to the toll paying boundary crossing trips. However this increase in consumers’ surplus with jurisdiction size does not offset the loss due to a leveling off of revenue and a steady increase in cost. Thus per capita welfare declines despite the increase in total welfare. This graph indicates that small jurisdictions have a higher per capita payoff from tolls than larger jurisdictions, and thus a greater incentive to toll. This policy is profitable for jurisdictions of through almost 50 km in length, similar in size to a metropolitan area.

A key point to note is that welfare remains positive despite the fact that costs exceed revenue. This indicates that a mixed financing system of taxes and tolls maximizes welfare even when cordon tolls cannot be relied on to finance the roads alone. Total costs increase non-linearly with jurisdiction size because travel demand increases when the share of trips paying tolls decreases. It should be further noted that a toll policy always results in greater per capita welfare than a tax policy in the All-Tax Environment. To see this compare the curve denoted “ $W(\text{Toll})/\text{Size}$ ” with “ $W(\text{Tax})/\text{Size}$ ”. However, the differences get smaller as jurisdiction size gets larger.

Figure 6 illustrates the welfare over a range of jurisdiction sizes. The curve denoted “ $W(\text{Tax})/\text{Size}$ ” illustrates the case when Jurisdiction J_0 imposes taxes. While welfare rises with jurisdiction size, it does so at a decreasing rate. This is so because as jurisdictions get larger, the impact of other jurisdictions on local welfare diminishes, non-local policies affect a smaller proportion of locally-originating trips. So the curves in very large jurisdictions approach those of a tax-policy in an all-tax environment. It should be noted that welfare is negative at very small jurisdiction sizes, indicating the cost of the road outweighs its benefits.

The curves representing the welfare (W), consumers’ surplus (U), and profit (π) from a cordon toll policy in an all-cordon toll environment are also shown in Figure 6. Welfare is maximized at the largest jurisdictions here, rising continuously at a decreasing rate. The cordon toll policy tracks closely (though is always slightly greater than) the welfare resulting from a tax policy in this environment. Again welfare is negative at very small jurisdiction sizes, though higher than the tax policy case, suggesting a very high

finance externality. This scenario is profitable for jurisdictions over a limited range of jurisdiction sizes (from about 2 km through above 20 km). This peaked shape is due to the interaction of jurisdictions. Closely spaced tollbooths (found in small jurisdictions) have a much greater finance externality than a farther spacing, so demand for longer trips increases with spacing of tollbooths. However, profit declines for larger jurisdictions because revenue reaches a maximum while costs increase steadily. The welfare increase indicates that the rise in consumers' surplus outweighs the decrease in profit.

We can compare welfare over the range of jurisdiction sizes under the assumptions outlined above for the four strategies. The highest welfare is attained by imposing tolls when others impose taxes (all-tax environment), followed by imposing taxes in the all-tax environment, imposing tolls in the all-cordon toll environment, and imposing taxes in the all-cordon toll environment. In the larger jurisdictions (above 100 km in length) the welfare measurements are very close and almost independent of policy, however, imposing tolls always beats imposing taxes in either environment.

9. POLICY SELECTION

We can define four cases which describe a jurisdiction's possible strategic decision with regard to tax or toll policy:

1. Always Tax
2. Always Toll
3. Mixed (Do the opposite): Toll when everyone else is taxing, but Tax when everyone else is tolling
4. Mixed (Do the same): Tax when everyone else is taxing, but Toll when everyone else is tolling

Three of the four cases appear in the range of data examined below. Clearly the mixed solution "Do the opposite" is not, in and of itself, stable. If every jurisdiction is supposed to toll when others tax and tax when others toll, they will flail about in their policy selection. Rather a certain fraction of jurisdictions invoking one policy and the rest the other is more likely to be stable. The solution "Do the same" has at least two stable points, everyone taxing and everyone tolling, though which will be achieved depends on the evolution of the system, or perhaps on which gives higher welfare, depending on the behavior ascribed to the decision-makers and the process they employ.

We compute the fixed collection costs (C_{CF}) necessary to equate tax and toll policies under each environment. We need to find the fixed collection cost where welfare from tolling equals the welfare from taxing under the all-tax environment ($W_{tc} = W_{cc}$), and

the all-cordon toll environment ($W_{tt} = W_{ct}$). The equations in the case of the all-tax environment are given below (the equations in the all-cordon toll environment are similar):

$$\begin{aligned} W_{tc} + C_{CF} - C'_{CF} - W_{cc} &= 0 \\ C'_{CF} &= W_{tc} + C_{CF} - W_{cc} \end{aligned} \quad (11)$$

Figure 7 illustrates policy selection as fixed collections costs (per jurisdiction) and jurisdiction size vary. The figure shows that when fixed collection costs (from both the entry and exit toll booths) are high, jurisdictions should (in the interest of maximizing local welfare as defined here) *always tax* (regardless of what other jurisdictions do); when they are low, jurisdictions should *always toll*. Small jurisdictions, basically below 20 km in length, also have a large mixed solution for collection costs between \$100 and \$700 of *tax when other jurisdictions toll and toll when other jurisdictions tax*. The exact threshold where a policy shifts from being "always tax" to "tax when other jurisdictions toll and toll when other jurisdictions tax" varies with jurisdiction size. Two factors influence the location of this threshold. First, large jurisdictions spread fixed collection costs over a larger number of users. Second, small jurisdictions suffer a finance externality from other jurisdictions' tolls. There is a large range of values where the policy choice depends on the behavior of neighboring jurisdictions. Because tolling when everyone else taxes is not a stable equilibrium among identical jurisdictions, the mixed region suggests a mixed set of policies. Thus we have a situation where some proportion of jurisdictions tax and others toll in order to arrive at a stable equilibrium. We might also consider the shape of those areas. As jurisdiction size gets larger, the *always tax* area gets larger (takes effect at a lower fixed collection cost). The *always toll* region is relatively flat with jurisdiction size.

Policy choice as a function of variable collection costs is more complicated than as a function of fixed collection costs. Though tolls are independent of fixed collection costs, they depend on variable collection costs (and thus the variable q). Similar to the case of fixed collection costs, we need to find the variable cost coefficient (q') where welfare from tolling equals the welfare from taxing under both the all-tax and the all-cordon toll environment. For the all-tax environment, we have the following equations (the equations in the all-cordon toll environment are similar):

$$\begin{aligned}
\text{Let } C'_{CV} &= \frac{C_{CV}}{q} \\
W_{tc} + C_{CV} - C'_{CV} * q - W_{cc} &= 0 \\
q &= \frac{W_{tc} + C_{CV} - W_{cc}}{C'_{CV}}
\end{aligned} \tag{12}$$

Figure 8 shows the results where the welfare is calculated using the collection costs (q') which are consistent with the welfare maximizing tolls resulting from those collection costs. This calculation uses a recursion procedure to solve for both the welfare maximizing toll and the collection cost necessary to equalize the welfare from toll and from tax policies. It was not possible to obtain solutions in all cases, for instance, the points not shown on the figure. Examining the variable collection costs, we see that both curves are downward sloping. The lower the collection cost, the more likely a jurisdiction will toll, but the larger the jurisdiction size, the more likely it will tax. Our estimate of collection costs using conventional technology (estimated from California data) falls below the two results for all values of jurisdiction size, indicating that with the baseline scenario we expect the always toll solution, consistent with the one-shot game described in a previous section and consistent with the results for fixed collection costs. We find in practice that given equal collection costs, large states are less likely to toll than small states. This model result confirms our expectation that the two curves be downward sloping (the *always toll* area gets smaller and the *always tax* area gets larger as jurisdiction size increases) for both fixed and variable costs. The figure also shows the tolls consistent with the collection costs necessary to equate the welfare from tax and toll policies. It is interesting that the tolls are decreasing with jurisdiction size. This is at odds with the previous cases, where the coefficient for variable collection costs was not varied.

10. GOVERNMENT HIERARCHY

We can use the information we have generated to examine the consequences of larger or smaller governments. This is analogous to the serial monopolist problem, where the toll paid by a traveler using two serial monopolists over an area is more than if the area were under the control of a single monopolist.

Table 6 summarizes the welfare maximizing tolls from the analysis above of a toll policy in an all-cordon toll environment as jurisdiction size varies. For instance traveling 20 km through two 10 km jurisdictions, our analysis finds that a traveler pays four tolls of

\$0.65 (\$2.60). However, traveling 20 km through one 20 km jurisdiction, the traveler only pays two tolls of \$0.68 (\$1.36), much less.

Table 6 also summarizes the welfare for the case of a toll policy in an all-cordon toll environment. Here, more total welfare is generated for one government of 20 km than the sum of two 10 km governments. For this case, the consolidation of jurisdictions eliminates a finance externality, and thus increases welfare. However, under a different scenario (for instance lower collection costs), the welfare gains from a reduction in the finance externality may be outweighed by the efficiency gains from a reduction in free riders associated with closer toll-booths.

Here we assumed no diseconomies of scale associated with higher levels of government. However two particular issues should be kept in mind. First, higher government levels have a broader span of control, so for the same number of managers, each area gets less attention, or more levels of management must be appointed. This can be more costly than local governance. Second, logrolling in a political environment can be a problem when resources need to be allocated centrally to numerous constituencies. Logrolling can lead to inefficient investments as the incentives for efficiently managing other people's money are less than for managing one's own.

11. STABILITY AND UNIQUENESS OF TOLLS

We now consider how a jurisdiction's payoff maximizing toll depends upon its neighbors' tolls. Figure 9 constructs reaction curves for four cases: two objectives (profit maximizing (π^*) and welfare maximizing (W^*)) each solved at two different jurisdiction sizes (1 km and 10 km). The profit maximization objective is introduced for comparison purposes. These two sizes were chosen as they are the most likely to show differences. As jurisdiction sizes get large, the interaction between the tolls at one cordon and the next gets small. All other conditions are identical to those assumed in the previous sections. Under welfare maximization, the tolls in jurisdiction J_0 are almost completely (but not entirely) independent of the tolls in other jurisdictions. To the extent they interact, welfare maximizing tolls are negatively correlated (a rise in one begets a fall in another). Under profit maximization, the tolling policies are more dependent and positively correlated (a rise in one begets a rise in another). Furthermore, profits are more sensitive in small jurisdictions than large. The profit maximizing toll is higher than the welfare maximizing toll, all else equal.

The increased sensitivity to other jurisdiction's tolls of the profit maximizing objective in comparison with welfare maximizing objective should not be surprising. The welfare maximizing objective depends on the consumers' surplus of local residents, while the profit maximizing objective doesn't. Rises in the price level will increase profits to a point, but will always reduce consumers' surplus, so welfare maximization is subject to more offsetting factors than profit maximization.

The diagonal (45°) line intersects each curve at the Nash equilibrium point when all jurisdictions are identical in all aspects, including the use of a toll policy. Thus in the case of welfare maximization, the Nash equilibrium toll is \$0.59 for jurisdiction sizes of 1 km, while it is about \$0.65 for jurisdictions of 10 km in length. Similarly, for profit maximization, Nash equilibrium tolls are \$0.98 and \$1.12.

The welfare maximizing tolls presented in earlier sections under an all-cordon toll environment were computed with an algorithm which began by assuming that other jurisdictions had a given "seed" toll ($r_t^{i=0}$), computing the toll for J_0 , and then updating the tolls assessed by other jurisdictions. This process was iterated to convergence. We want to confirm that the initial seed toll does not affect the result. Figure 10 illustrates this for one set of assumptions (jurisdiction size = 1 km, welfare maximizing, non-cooperative toll-setting, empirical coefficients). It shows that for four different seeds (-\$1, \$0, \$1, and \$10), they all converge after three iterations to the same value (within 6 decimal places here). This result has been repeated for numerous other cases (not shown). We have thus corroborated experimentally that a unique solution is independent of initial conditions.

12. SENSITIVITY TO MODEL COEFFICIENTS

This section examines the sensitivity of the basic welfare measures to model coefficients, under standardized assumptions of a local welfare maximization objective, a 10 km jurisdiction size, a toll policy and an all-cordon toll environment where jurisdictions employ non-cooperative toll-setting. Each of the variables is examined over a range of coefficient values approximately centered upon the default values developed in Table 5. Table 7 summarizes the elasticity about the mean for each of the key variables. The elasticity is calculated by calculating the change in welfare (revenue, cost, consumers' surplus, profit) over the difference between two values of the variable in question, where one value is the mean, and the other is a one percent increment on the mean.

First, the sensitivity of demand to cost in general, the variable alpha (α), is evaluated over a range of values (recall the default value of -1.00). Clearly, as expected

from inspection of the model as alpha increases in absolute value, welfare, revenue, cost, and consumers' surplus drop.

Second, the density of trip origins, the variable delta (d), is evaluated with a default value of 180. As the density of trips increase, total welfare, consumers' surplus, revenue, and cost rise. Under the model, the more trips, the higher the welfare, because there is no offsetting congestion factor.

Third, the private fixed costs per trip, the elasticity for the variable zeta (Z), is computed with a default value of 1.23. As might be expected, as this cost rises, welfare, costs, revenue and consumers' surplus drop.

The fourth variable psi (y) which is the sensitivity of demand to trip length $|y-x|$ has a default value of 0.018. As sensitivity increases, welfare, revenue, cost and consumers' surplus decrease.

The fifth variable gamma (g) is the coefficient on jurisdiction size (in \$/km) in the cost equation. Recall that we have set fixed costs equal to zero in the analyses in previous sections. As gamma rises, costs rise (by definition) and welfare falls continuously. However, only as costs approach \$100 per linear kilometer per hour do they noticeably influence the welfare indicators. Revenue and consumers' surplus remain unchanged, because this variable does not affect tolls.

The sixth variable, phi (f), is the coefficient which reflects the variable cost to the jurisdiction as a function of vehicle flow, with a default value is \$0.018/vkt. As the variable cost increases, per capita cost increases (again, by definition), and welfare, consumers' surplus, and profit fall. Interestingly, revenue peaks at a value of about \$0.10/vkt, as costs rise, so do tolls, but above a certain point, the tolls are sufficiently high to drive away the significant amount of toll-paying (inter-jurisdictional) traffic.

Finally, we consider theta (q), the variable cost of toll collection as a function of flow past the toll-booth, with a default value of \$0.08/trip. As this coefficient increases, welfare and consumers' surplus fall. Costs rise up to \$0.20, and then fall, as the increase in welfare maximizing tolls drives away more traffic (and thus reduces costs) than the increase in variable collection costs increases costs.

13. CONCLUSIONS

While many jurisdictions do not actually consider their choice of revenue mechanism, they probably should. Changes in technology and funding sources suggest that a reconsideration of the standard financing mechanisms is warranted. The purpose of this research was to develop a model to explain whether jurisdictions choose to tax or to toll as a function of the length of trips, the size of the jurisdiction, the transactions costs of collecting revenue, demand, and the cost of providing infrastructure. In short the intent was to help frame the analysis of how jurisdictions have historically considered, albeit in a more complex and less mathematical environment, and may in the future consider, the choice of revenue mechanism.

Large jurisdictions are more likely to impose taxes or a mixed financing policy than only cordon tolls because cordon tolls raise insufficient revenues to cover costs, as revenue levels off above a certain point. Similarly, the higher the cost of collecting tolls, the less likely tolls will be the preferred revenue mechanism. The welfare maximizing toll may not fully recover costs, and still require subsidy. The maximum welfare from taxes may exceed those of tolls under certain circumstances, depending on model parameters, however if jurisdictions are sufficiently small, demand sufficiently higher, and collection costs relatively low, then tolls will be preferred.

The gains to a jurisdiction of imposing tolls exceeds the gains from taxes under certain circumstances. The gains come foremost from residents of other jurisdictions. This problem, a finance externality, is well known in certain cases, for instance the reliance by local governments on some mix of sales, income, and property taxes, each of which are borne by a different set of people, not all of whom are local.

Congestion pricing has long been a goal of transport economists, who argue that it will result in more efficient use of resources. The path for implementing such a system has been strewn with political potholes, pricing inevitably creates winners and losers. An alternative approach, one which would create the local winners necessary to implement road pricing, is required before congestion pricing can be expected to become widespread. This research suggests one approach, one that would decentralize the decision about whether to tax or toll before attempting to impose road pricing. Road pricing is a necessary prerequisite to congestion pricing, which once tolls are in place, is not nearly as difficult problem as placing tolls on untolled roads in the first place. And tolls are a rational financing mechanism for a sufficiently small jurisdiction, particularly with the advent of electronic toll collection systems.

The prospects for future success of toll roads depend on several factors, including the relative centralization of control of the highway sector, and the transaction costs of

collecting revenue. Factors that would be conducive to a return to turnpikes are a reduction in collection costs and a decentralization of authority. Should the governance become more decentralized, and collection costs continue to drop, tolls could return to prominence as the preferred means of financing roads for both local and intercity travel.

The analysis presented herein is a simplified representation of reality. Future specific extensions include: congestion and travel times as a function of use, public vs. private ownership, competition in the network between alternative routes, multiple owners and different numbers of owners of alternative routes, degrees of vertical and horizontal integration, scale, scope, and sequence economies, and heterogeneous users with different values of time which might lead to a differentiated network.

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Table 1 General Trip Classification

	Section of	Destination	(y)
	S_-	J_0	S_+
Section of Origin (x)	S_-	G_{--}	G_{-+}
	J_0	G_{0-}	G_{0+}
	S_+	G_{+-}	G_{++}

note: shading indicates trip classes of interest on a one-way road.

0 indicates jurisdiction J_0 ; -, + indicates jurisdiction in S_- , S_+ respectively

Table 2 Model Components: General Tax and Cordon Toll

Y	Description	User Group	b_{YG}
C_{CV}	Variable Collection Costs	G_{00} $G_{0+} G_{-0}$ G_{-+}	0 qd qd
C_{CF}	Fixed Collection Costs	All	kK_i
C_r	Variable Network Costs	G_{00} $G_{0+} G_{-0}$ G_{-+}	$jd w^3 v^3 u(t \ln(t) + \ln(t) - 2t + 2)$ $\frac{-jd wv(1+t)(-1 + \ln(t))}{(1-t)}$ $jd (b - a)$
C_s	Fixed Network Costs	All	$g_w v \ln(t)$
R	Toll Revenue	G_{00} G_{-0}, G_{0+} G_{-+}	0 $r_t d$ $2r_t d$
U	Consumers' Surplus	G_{00} G_{0+} $G_{-0} G_{-+}$	$dw^3 v^2 u(1 - t + \ln(t))$ $\frac{-d}{a}$ 0

where: $v = \frac{1}{y}$, $w = \frac{1}{a}$, $t = e^{ay|b-a|}$, $u = e^{az}$, $y = n + \frac{V_T}{S_f}$,

equation form: $Y = g_G b_{YG}$

note: Fixed costs (C_{CF} , C_s) do not vary with demand, and thus $g_G = 1$ for fixed costs.

Table 3 Mathematical Solution Associated with User Class (g_G) for General Tax and Cordon Toll Policies in J_0 and Environment

Class	g_G
G_{00}	$g_{00} = 1$
G_{0+}	$g_{0+} = w^2 v^2 \frac{ue^{a(r_t + r_l)}(1-t)^2}{(1 - te^{a2r_l})}$
G_{-+}	$g_{-+} = w^2 v^2 \frac{ue^{a2(r_t + r_l)}(-1-t^2)}{\hat{A} - te^{a2(r_l)}^2}$

Table 4 Rate of Toll Under Various Policies

Environment		Policy	
J_0		tax	toll
	tax	$r_t^{cc} = 0$ $r_l^{cc} = 0$	$r_t^{ct} = 0$ $r_l^{ct} = r_t^{*tt}$
	toll	$r_t^{tc} = r_t^{*tc}$ $r_l^{tc} = 0$	$r_t^{tt} = r_t^{*tt}$ $r_l^{tt} = r_l^{*tt}$

note: r_t = toll set by J_0 , r_l = toll set by all other jurisdictions

Table 5 Baseline Scenario: Empirical Values of Model Coefficients

Variable	Description	Value
zeta (z)	fixed cost of vehicle ownership (\$/veh-trip)	1.23
psi (y)	$n + V_T / S_F$ variable cost of travel (\$/vkt)	0.15
alpha (a)	coefficient relating demand to price	-1
delta (d)	total demand multiplier (trips)	180
kappa (k)	fixed collection cost (\$/toll-booth/hour)	\$90
gamma (g)	fixed network cost (\$/km)	0
theta (q)	variable collection cost (\$/vehicle)	0.08
phi (f)	variable network cost (\$/vkt)	0.018

ref.: Levinson (1998)

Table 6 Comparison of Tolls and Welfare for Different Jurisdiction Sizes

Jurisdiction Size (km)	Tolls	Welfare
10	\$0.65	2367
2x10	\$1.30	4734
20	\$0.68	5451

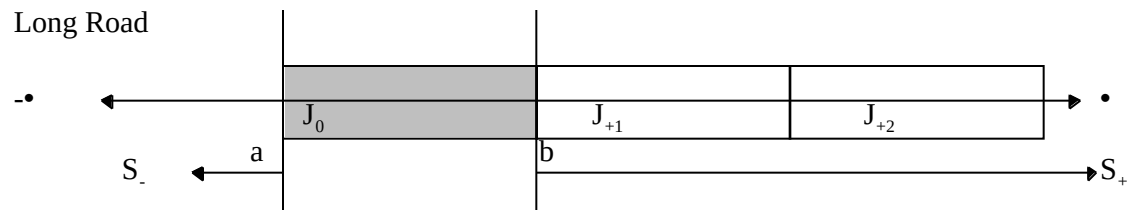
source: author's analysis of toll policy for J_0 and all-cordon toll environment

Table 7 Elasticity About Mean

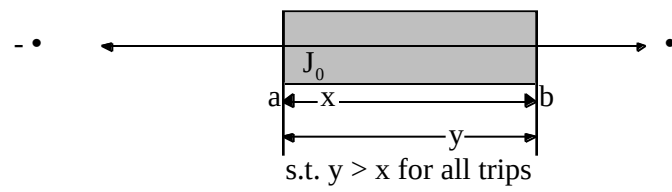
Variable		Welfare	Revenue	Cost	Cons. Surplus	Profit
Alpha	sensitivity of demand to cost	-3.070	-3.437	-1.758	-2.785	-5.320
Delta	density of trip origins	1.038	1.000	0.699	1.000	1.337
Zeta	private fixed costs	-0.424	-1.150	-0.498	-0.239	-1.879
Psi	sensitivity of demand to trip length	-0.094	-0.143	-0.119	-0.065	-0.156
Gamma	fixed costs associated with network length	-0.044	0.000	0.250	0.000	-0.596
Phi	variable cost associated with network use	-0.049	-0.028	0.277	-0.017	-0.251
Theta	variable collection costs	-0.016	-0.013	0.053	-0.009	-0.065

Note: Elasticity of Welfare (Revenue, Cost, Consumer's Surplus, Profit) About Mean of Alpha (Delta, Zeta, Psi, Gamma, Phi, Theta)

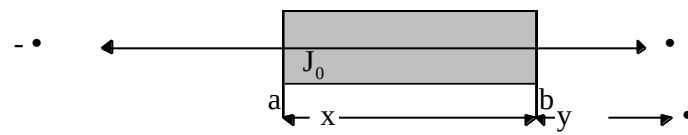
Figure 1 One-Way, Long Road and Classes of Trips



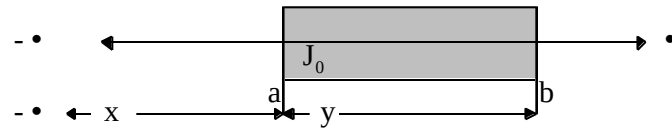
Class G_{00} Trips



Class G_{0+} Trips



Class G_{-0} Trips



Class G_{-+} Trips

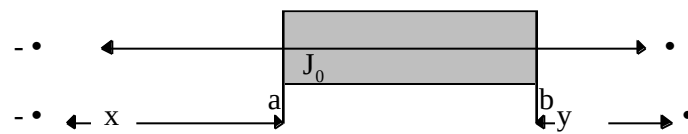


Figure 2 Distance Decay and Tolls in All Cordon Toll Environment.

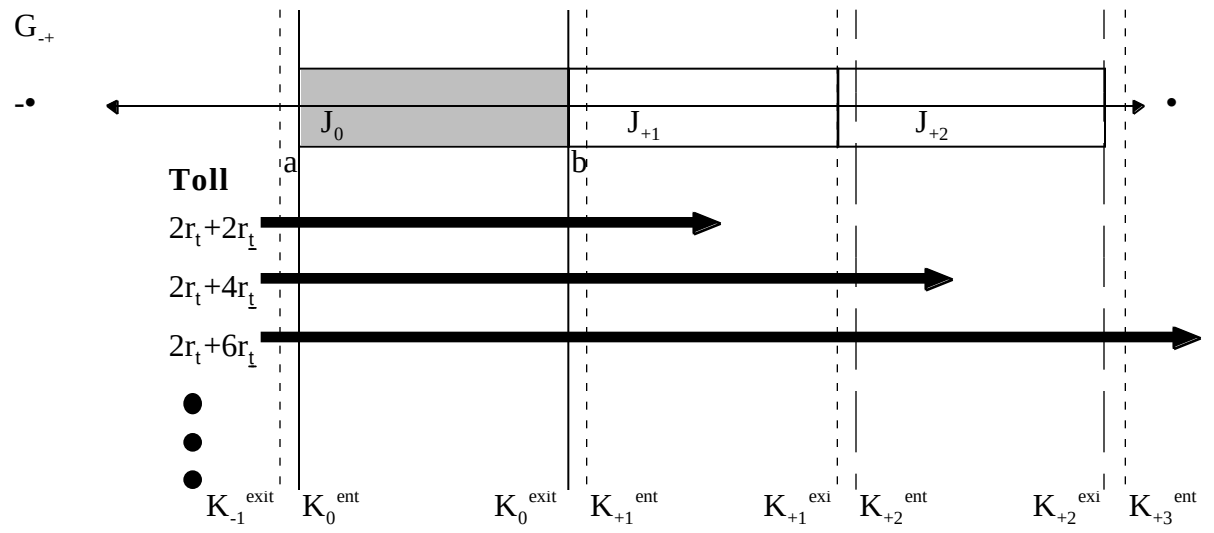


Figure 3 Welfare in J_0 as a Function of J_0 Toll in an All-Tax Environment

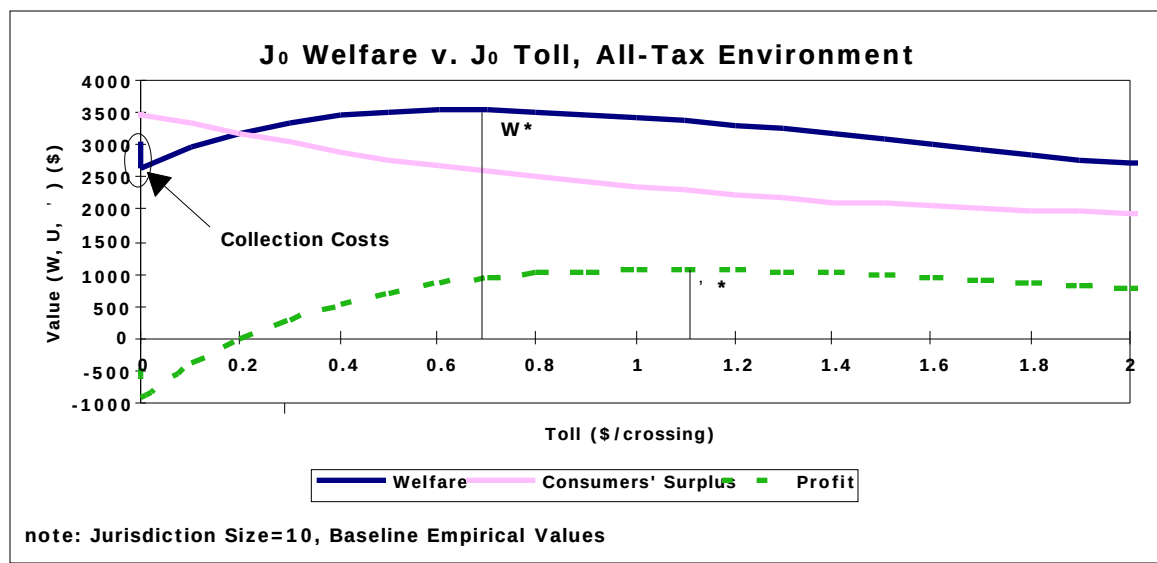


Figure 4 Welfare in J_0 as a Function of J_0 Toll in an All-Cordon Toll Environment

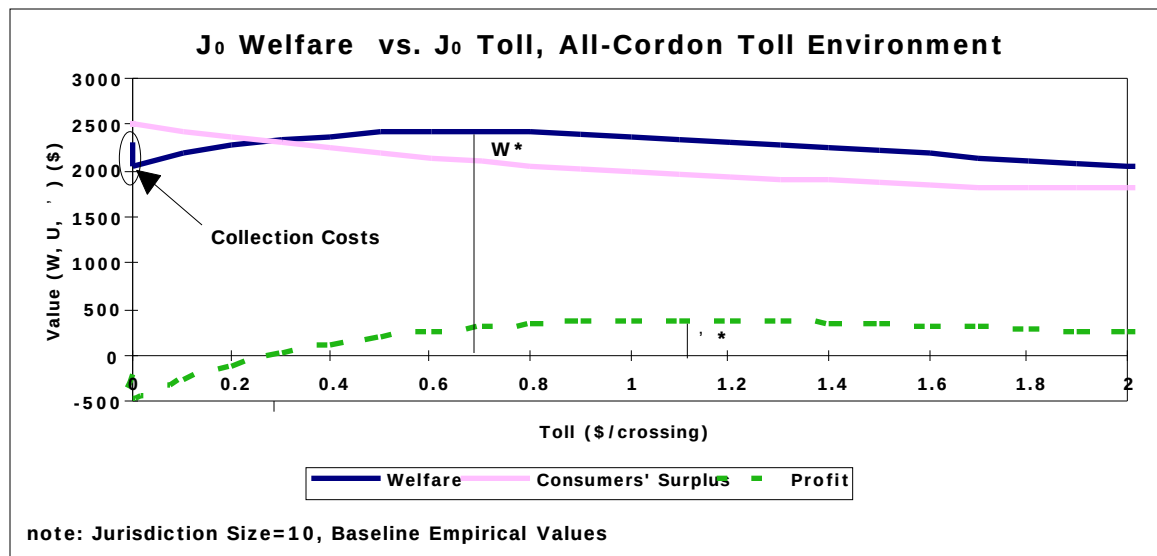


Figure 5 Welfare in J_0 at Welfare Maximizing Tolls as Jurisdiction Size Varies in an All-Tax Environment

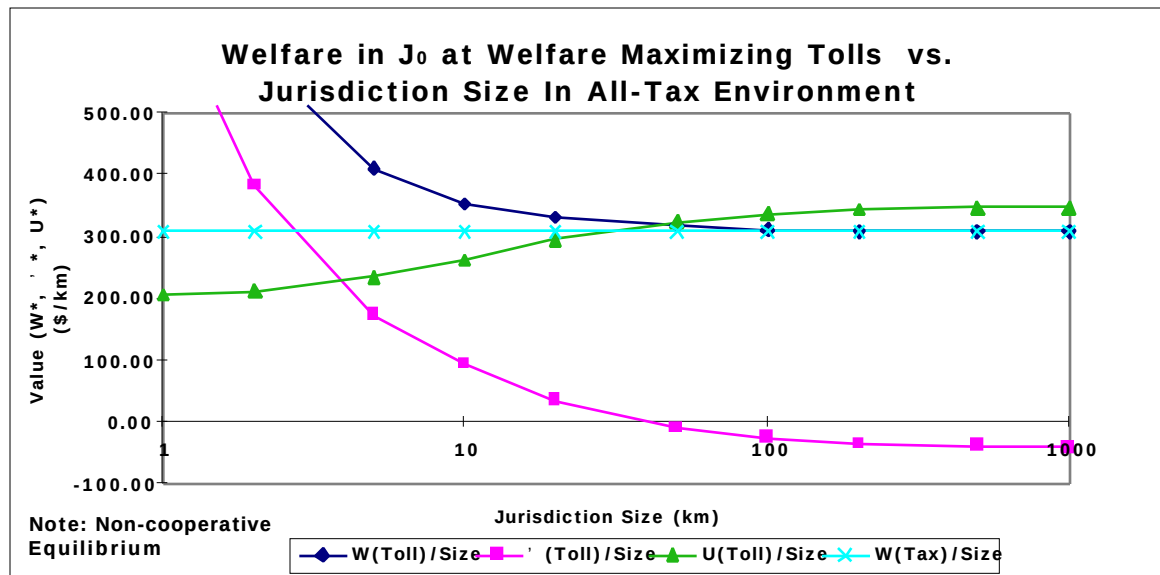


Figure 6 Welfare in J_0 at Welfare Maximizing Tolls as Jurisdiction Size Varies in an All-Cordon Toll Environment

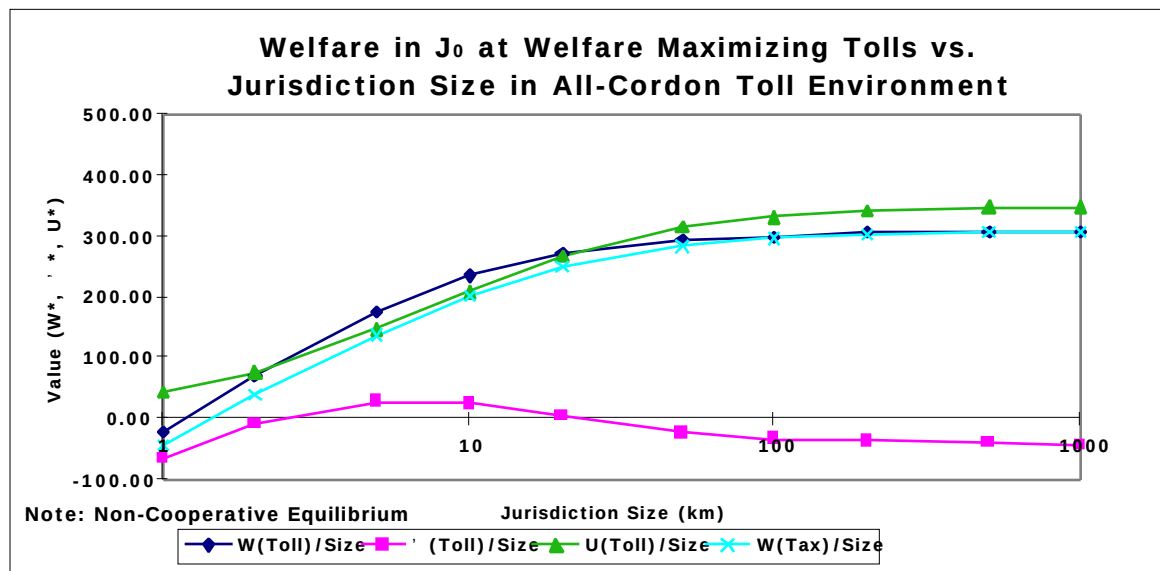


Figure 7 Policy Choice as a Function of Fixed Collection Costs and Jurisdiction Size

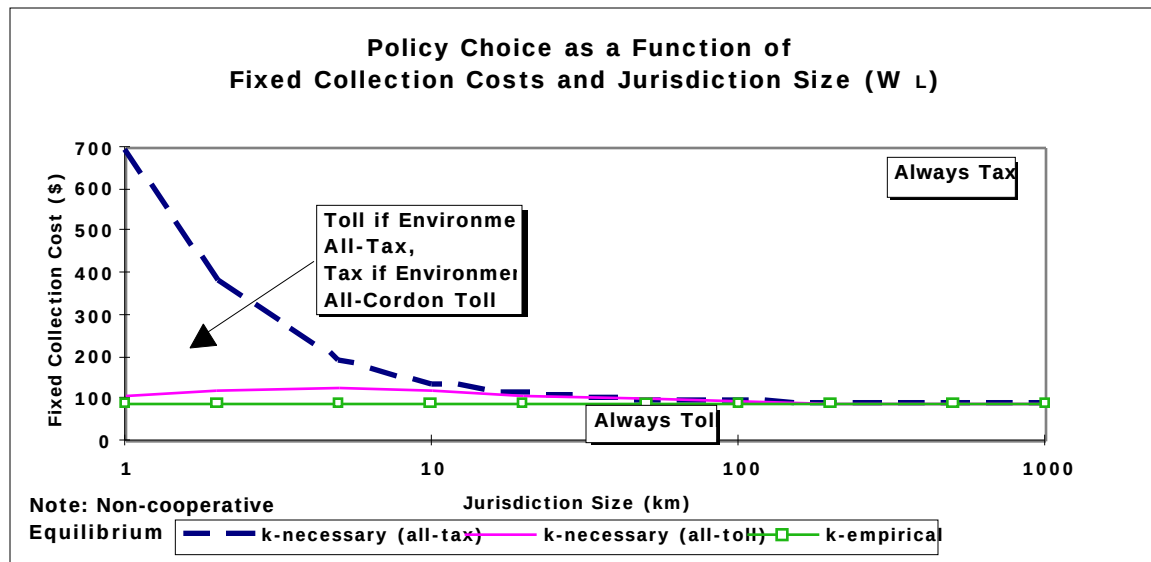


Figure 8 Policy Choice as a Function of Variable Collection Costs and Jurisdiction Size

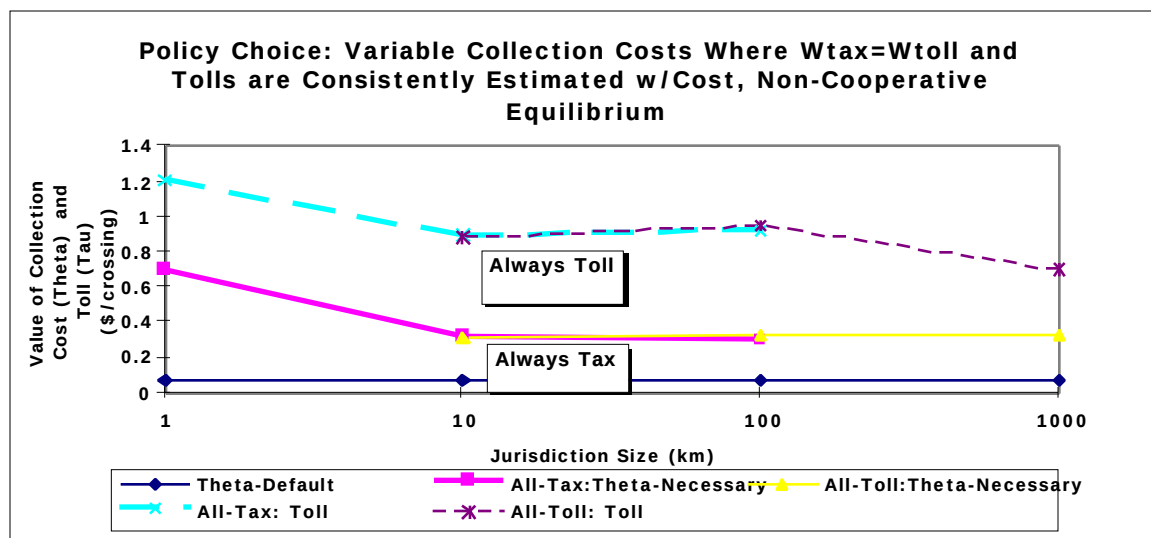


Figure 9 Reaction Curves: Best J_0 Toll as Tolls Vary in an All-Cordon Toll Environment

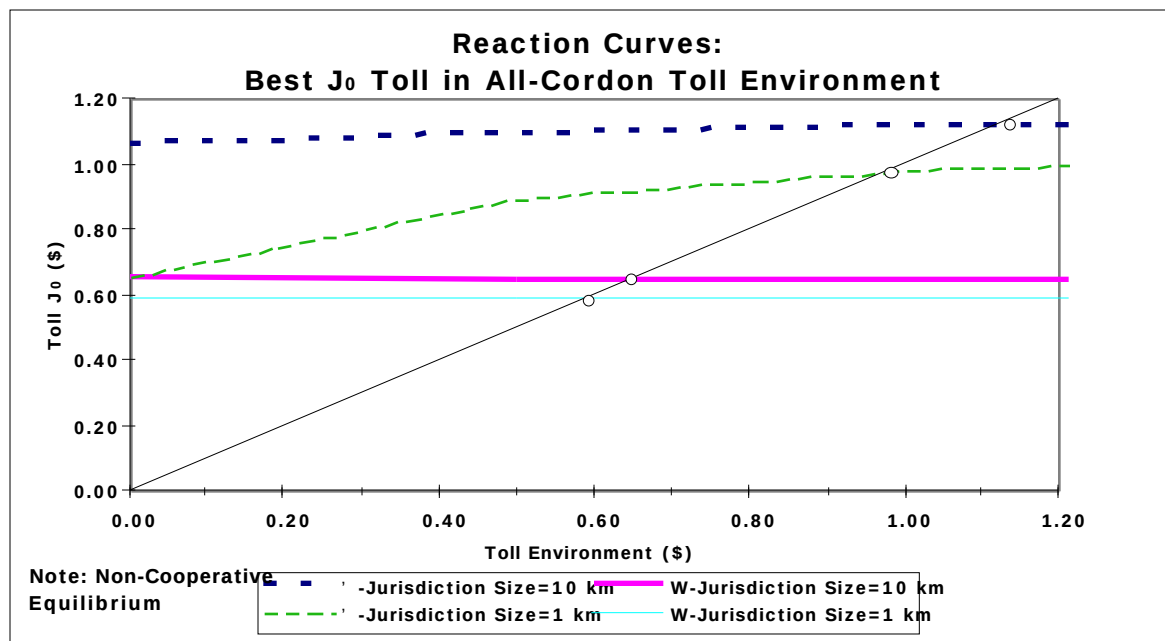


Figure 10 Uniqueness: Non-Cooperative Welfare Maximizing J_0 Toll as Initial Toll for Other Jurisdictions Varies in All-Cordon Toll Environment

